Challenge: Obstacle Avoidance

Points: 250 • Estimated Time: 90 Minutes

Challenge Scope

The robot must detect and drive around 3 randomly placed cardboard obstacles and reach the finish zone.

Instructions & Engineering Hints

• **Objective:** Program and equip your robot to autonomously navigate from the starting zone, successfully detect and drive around 3 randomly placed cardboard obstacles, and finally cross into the finish zone without manual intervention.

• **Sensor Placement Tips:** Consider where your ultrasonic or infrared sensors are mounted. If they are placed too low, they might miss smaller variations in the obstacles; if placed too high, they might not detect them at all. Think about expanding your robot's field of view by angling sensors slightly outward or using multiple sensors for side-clearance checks.

• **Strategic Approaches:** Instead of just stopping when an obstacle is detected, design a state machine or a modular logic loop (e.g., Forward -> Detect -> Reverse/Turn -> Bypass -> Re-orient). Utilizing encoder counts or timed turns can help your robot make predictable, geometric maneuvers around each box.

• **Structural Considerations:** Ensure your robot's chassis is stable and that cables are securely tucked in. Hitting an obstacle at speed might dislodge poorly mounted sensors, causing erratic behavior mid-run.